Quantitative Trading: Theory and Practice
Advanced Course – Lecture 3

Backtesting Discipline, Trading Costs, and Robustness Checks

Dr. Lianghai Xiao

Jinan University

Lecture 3 Positioning

- In Lecture 1, we studied factor construction and factor testing.
- In Lecture 2, we studied multi-factor combination and portfolio construction.
- In this lecture, we ask a more critical question:

  Is a profitable backtest necessarily a credible strategy?

- The main objective is to understand how to evaluate a strategy in a disciplined and realistic way.

Learning Objectives

By the end of this lecture, students should be able to:

- diagnose why a profitable backtest may still be unreliable, especially when time alignment, universe definition, or sample selection is flawed;
- incorporate transaction costs, turnover, and implementation frictions into strategy evaluation rather than relying only on gross returns;
- design robustness checks through parameter sensitivity, subperiod analysis, and cost stress testing to distinguish stable evidence from overfitting;
- interpret and report backtest results responsibly, with clear discussion of assumptions, limitations, and practical credibility.

Why a Good Backtest May Still Be Wrong

A profitable backtest is not automatically a trustworthy strategy. Typical failure channels:

- **Spurious statistical relation:** the apparent edge may reflect nonstationarity, data snooping, or an unrepresentative sample rather than a stable effect.
- **Data and timing errors:** signals, returns, or universe filters may be misaligned with the information actually available at the decision time. Survivorship and hindsight universe selection.
- **Unrealistic implementation:** transaction costs, slippage, market impact, or trading constraints may be ignored.
- **Weak economic robustness:** the apparent edge may be statistically significant but lack a credible economic foundation.

Key message

A backtest can fail not only because the statistical relation is spurious, but also because the test is mismeasured, unrealistic, or economically fragile.

Spurious Regression

An apparent statistical relationship may look significant in sample, even though it does not reflect a stable predictive or economic mechanism.

- **Nonstationarity:** the data-generating process changes over time.
- **Data snooping:** too many trials are run before only the winner is reported.
- **Limited samples and regime dependence:** the result is dominated by one exceptional subperiod or one favorable market environment.

Practical interpretation

These problems can produce a high Sharpe ratio in sample without implying a stable and repeatable trading opportunity out of sample.

Source 1: Nonstationarity

Spurious regression often appears when the relationship estimated in the sample is not stable over time.

- market microstructure, regulation, and investor composition can change;
- volatility and liquidity regimes can shift;
- a factor that worked in one regime may weaken or reverse in another.

Practical example

A small-cap strategy performs very well in 2022–2023, when liquidity is tight and retail trading is active. If the same strategy performs poorly from the beginning of 2024 under different market leadership, then the original fit may reflect a regime-specific pattern rather than a stable effect.

Lesson: a high full-sample result can be misleading if it averages over structurally different periods.

How to Deal with Nonstationarity

Nonstationarity cannot be eliminated, but it can be handled more carefully in research design.

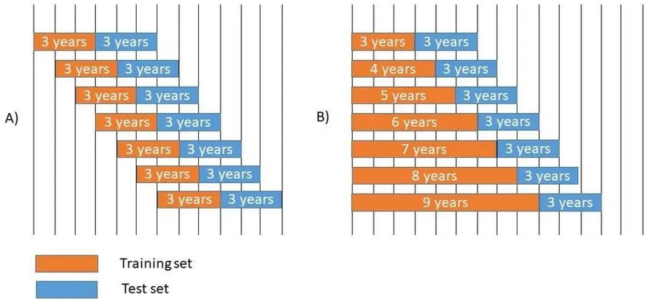
- test the strategy separately across different subperiods and market regimes;
- use rolling or expanding windows rather than relying only on one full-sample estimate;
- prefer signals with clear economic logic instead of purely empirical fit;
- monitor whether IC, turnover, and performance decay materially over time;
- update model parameters cautiously when evidence suggests structural change.

Practical example

For a reversal strategy, instead of estimating its effectiveness once on 2016–2025, we can examine rolling 24-month ICs, compare bull and bear markets, and check whether the signal weakens after transaction costs in later years. If the effect survives across these settings, the evidence is more credible.

Key idea: the goal is not to assume stability, but to test how sensitive the strategy is when the market environment changes.

Rolling Windows vs. Expanding Windows



How to use them in practice

If we estimate a factor IC, a rolling 24-month window asks whether the signal still works *recently*, while an expanding window asks whether the signal remains supported by the *entire history so far*.

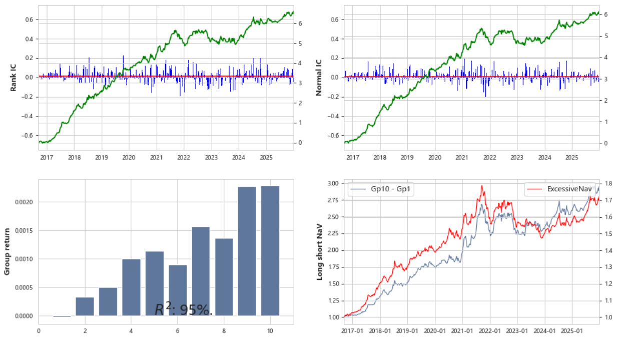
Rolling window

- uses a fixed-length recent sample;
- drops old observations as new data arrive;
- reacts faster to regime change.

Expanding window

- starts from an initial sample and keeps adding new data;
- uses all available history up to the current date;
- is more stable, but slower to adapt.

Monitoring IC, Turnover, and Performance Decay



Interpretation

- **IC predictive power**: if IC decays first, the signal itself is weakening;
- **Long–short spread, Sharpe ratio**: if gross returns survive but net returns decay, the edge may no longer be tradable.

Example: SUE factor

This figure shows that the SUE factor's IC becomes insignificant after 2022. That is a warning sign of nonstationarity: even if the factor worked earlier in the sample, its predictive content may have weakened materially in the more recent regime.

Source 3: Data Snooping

Data snooping means repeatedly searching until a strong result is found, even if the true predictive content is weak.

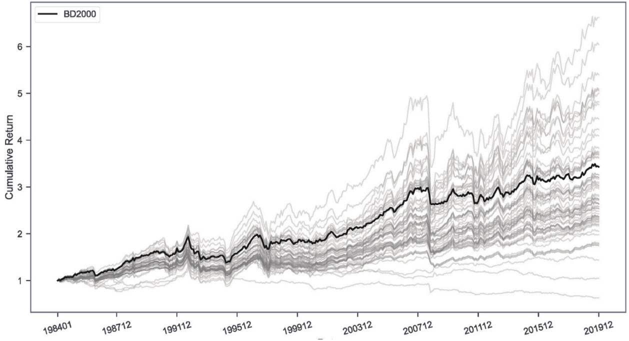
- many factors may be tested and only the best one retained;
- parameter grids may be searched aggressively;
- different universes, rebalancing rules, and filters may be tried until the backtest looks attractive.

Practical example

An analyst tests 80 factor formulas, 12 lookback windows, and 5 rebalancing frequencies. One version achieves a Sharpe ratio above 2.0, so it is presented as the "strategy." But if thousands of combinations were tried, the winner may simply be the luckiest draw from noise.

Lesson: the more strategies you search over, the less convincing the best in-sample result becomes.

Data Snooping in Machine Learning Strategies



Machine learning often makes data snooping more severe because the search space becomes extremely large.

- many features can be tried;
- many labels, architectures, and hyperparameters can be searched;
- many candidate strategies can be generated automatically.

Machine learning amplifies data snooping in many cases.

If we construct 20,000 candidate strategies and then keep only the best one, it is very likely that some "extraordinary" performers are simply the luckiest ones in sample.

How to Reduce Data Snooping

Data snooping cannot be eliminated completely, but it can be reduced by stricter research discipline.

- prefer simpler models when extra complexity adds little robust improvement;
- require economic intuition, not only backtest ranking, before keeping a strategy;
- separate the data into training, validation, and final test periods;
- avoid repeatedly reusing the same test set for model selection;
- check whether the selected strategy survives in new periods, new universes, and higher cost assumptions.

Key principle

The goal is not to find the most impressive in-sample strategy, but to find a strategy that still works after honest selection and out-of-sample evaluation.

Source 4: Limited Samples and Regime Dependence

A strategy may look convincing not because it is broadly robust, but because the sample is too limited or too concentrated in one favourable market regime.

- one crisis, rebound, or style-dominant episode may account for most of the performance;
- the estimated Sharpe ratio may change sharply when the start date or end date moves slightly;
- a strategy can look strong simply because the sample omits adverse regimes;
- once the test window is extended, the effect may weaken, reverse, or become insignificant.

Practical example

A short-selling strategy appears highly profitable when tested mainly over 2019–2020, especially during the U.S. equity market crash and circuit-breaker episodes. However, when the sample is extended to calmer or rising markets, the same strategy performs poorly or even loses money. The original backtest looked strong because it was dominated by one unusually favorable regime for short positions.

Limited Samples vs. Data Snooping

Data snooping

- comes from testing too many ideas and keeping the winners;
- bias is created by the *research process*;
- even with a long sample, the selected strategy may still be a false positive.

Did we find this strategy because it is truly good, or because we searched over too many alternatives?

Limited sample / regime dependence

- comes from evaluating a strategy on too short or too special a historical window;
- bias is created by the *sample realization*;
- randomly chosen strategy looks strong if the period is unusually favourable.

Did this strategy work broadly, or did it only look good in one unusually supportive market environment?

Both can produce impressive backtests.

Data snooping is about *how many strategies we tried*. Limited-sample bias is about *which period we happened to observe*.

Data and Timing Errors

Backtests can look strong simply because the data pipeline does not respect what was truly known, when it was known, and how returns are measured.

- **Information leakage** uses data that were not available at the decision time.
- **Alignment and execution errors** mismatch signal time, trade time, and realized return.
- **Revised / backfilled data errors** use ex post corrected or backfilled data that were not available in the same form at the historical decision time.

Core principle

A valid backtest must replicate the chronological sequence:

$$\text{available information} \rightarrow \text{signal} \rightarrow \text{trade} \rightarrow \text{future return.}$$

Source 1: Information Leakage and Biased Sample Construction

Definition

A backtest can be misleading if it uses information that was not truly available at the decision time, or if the historical sample itself is constructed with hindsight.

- **Publication lag ignored:** accounting or analyst data are used at the report date instead of the actual public release date.
- **Revised or backfilled data:** cleaned historical datasets may contain values that were not observable in real time.
- **Future universe information:** index membership, delisting status, or industry labels may be taken from a later date.

Once unavailable information or hindsight sample selection enters the pipeline, the backtest no longer represents a feasible trading process.

Information Leakage and Biased Sample: Practical Examples

1. Using same-week return ranks

Suppose we trade on Wednesday but rank stocks by their *weekly* return. If that weekly return already includes Thursday or Friday prices, then the ranking uses information from the future.

2. Financial data before release

Suppose we form a signal at the end of March using 2025Q1 earnings, but those financial statements were published only in April. The backtest is implicitly assuming access to unreleased information.

3. Full-sample beta estimation

Suppose we use the whole 2015–2025 dataset to estimate each stock's beta, and then construct a factor for 2018. That beta estimate contains return information from years after 2018.

4. Future index constituents as universe

Suppose we backtest on the CSI 300 universe but assume that the constituent list is already known throughout history. The strategy benefits from future selection information using later index membership.

Universe Filtering and Sample Construction

Universe filtering is especially dangerous because both information leakage and biased sample construction can enter silently during data cleaning.

What should *not* be done

- remove delisted or suspended stocks retrospectively from the whole history;
- define listing-age, liquidity, or tradability filters using revised datasets;
- apply ‘dropna()‘ to the full DataFrame and then backtest on the remaining sample;
- keep only stocks with complete histories, which automatically favors survivors.

Why ‘dropna()‘ is risky

If a stock has missing values in later years and we drop that stock from the full panel at the beginning, then earlier observations are removed using future missingness information. This is a subtle form of look-ahead in universe construction.

Source 2: Alignment and Execution Errors

Even when all data are historical, the backtest can still be wrong if signal time, trading time, and return measurement are misaligned.

- **Signal-return mismatch:** factor values at time t are matched with returns that already overlap with the signal formation period.
- **Same-close execution:** the strategy uses the closing price to compute the signal and also assumes execution at that same close.
- **Holding-period mismatch:** the portfolio is rebalanced weekly, but the evaluation uses daily returns without matching the actual holding window.
- **Asynchronous merges:** signals, prices, and benchmarks from different frequencies or markets are joined without consistent timestamps.

Key point

The strategy may use only historical data and still be invalid if the order of *signal formation*, *execution*, and *return measurement* is not implemented correctly.

Alignment and Execution Errors: Practical Examples

1. Same-close execution

We compute the signal from the Friday closing price and also assume the trade is executed at that same Friday close. In practice, the signal is known only after that close.

2. Signal-return mismatch

If we rebalance weekly and evaluate the strategy using Friday's closing price, but the portfolio is actually traded and rebalanced during Friday, the realized execution price may differ from that closing price.

3. Holding-period mismatch

The portfolio is rebalanced every week, but performance is evaluated using daily signals and daily returns as if the strategy could adjust continuously. Reported performance no longer matches the actual rule.

4. Asynchronous merges

If we trade both the China and U.S. stock markets and merge their observations by calendar date. Even on the "same" date, the information from one may not have been available when trades were made in the other.

Source 3: Revised / Backfilled Data Errors

Some backtests fail not because the model is wrong, but because the dataset available *today* is cleaner or more complete than the dataset that was actually observable at that historical time.

- historical financial statements may later be corrected and replaced by revised values;
- vendor databases may backfill missing observations after the fact;
- revised classifications, cleaned fields, or patched time series may be written back into earlier dates;
- a backtest using such ex post data may unknowingly assume information quality that investors did not actually have in real time.

Practical example

On April 30, 2026, Wuliangye announced corrections to previously disclosed 2025 quarterly financial data. If a backtest later uses the revised 2025 numbers as if they had always been available in that form, then the strategy is being tested on ex post corrected data rather than on the information that investors actually observed at those earlier dates.

Unrealistic Implementation

A backtest may look profitable simply because it assumes a trading process that is too easy, too cheap, or too frictionless to achieve in practice.

- **Transaction costs:** commissions, fees, and taxes reduce net returns directly.
- **Slippage:** realized execution prices are often worse than observed quotes or closing prices.
- **Market impact:** large orders can move prices against the strategy itself.
- **Trading constraints:** liquidity limits, suspension, short-sale constraints, and order-size limits may prevent the intended trades.
- **Turnover exposure:** even a small per-trade cost becomes important when the strategy trades frequently.

Key point

A strategy should be judged not only by whether it predicts returns, but also by whether those returns can still survive realistic execution.

A Simple Cost-Aware Return Model

A basic net return model is:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \text{turnover}_t,$$

where

- c is a transaction cost parameter,
- turnover_t measures the amount traded at time t.

Equivalently, in portfolio notation,

$$\text{PnL}_t^{\text{net}} = w_{t-1}^\top \boldsymbol{r}_t^{\text{net}} < w_{t-1}^\top \boldsymbol{r}_t^{\text{gross}} = \text{PnL}_t^{\text{gross}}.$$

Since $\text{NAV}_t = \prod_{\tau=1}^{t} \left(1 + \text{PnL}_\tau\right)$, the final backtest based on net returns can differ substantially from the one obtained from gross returns.

Why Turnover Matters

Turnover is one of the most important practical diagnostics in portfolio research.

- high turnover means transaction costs are paid repeatedly rather than occasionally;
- high turnover usually increases slippage and market impact, especially in less liquid stocks;
- high turnover may indicate that the strategy is reacting to short-lived noise rather than persistent information;
- a strategy with lower gross return but much lower turnover can be more implementable after costs.

Interpretation

Suppose Strategy A earns a higher gross return than Strategy B, but it turns over the portfolio three times as fast. After commissions, slippage, and market impact, Strategy B may be the better real-world strategy.

Price Limits, Circuit Breakers, and Tradability Constraints

In some markets, a stock may be theoretically selected by the signal but not realistically tradable because of price limits or market-wide trading halts.

- when a stock is at its upper or lower price limit, execution may be impossible or highly uncertain;
- tradability filters must follow market-specific rules rather than one universal threshold;
- in A-shares, ST, main-board, and ChiNext stocks can face different daily price-limit bands;
- in the U.S., severe broad-based declines can trigger market-wide circuit breakers at 7%, 13%, and 20% (KOSPI in Korea, 8%, 15%, and 20%);
- realistic backtests should check both stock-level limits and market-level halts.

Practical implication

If we remove "limit-up" or "limit-down" days using one fixed rule, or ignore market-wide halts, we may assume trading was feasible when it was not. The backtest should apply the correct security-level and market-level rules rather than assuming frictionless execution.

Strategy Capacity and Trading Volume

A strategy may look strong in backtest but fail at larger scale because the market cannot absorb the required trades.

- capacity is often evaluated by comparing the strategy's average money position or trading demand with the stock's daily trading amount;
- illiquid stocks may contribute strong paper alpha but very limited deployable capital;
- a common practical rule is to cap the average money position or rebalance volume at roughly 5%–10% of a stocks average daily trading amount, with more conservative limits such as 1%–5% for less liquid names.

Practical implication

Suppose a portfolio wants to hold RMB 30 million in a stock whose average daily trading amount is only RMB 100 million. If the strategy already represents a large fraction of daily turnover, then rebalancing will likely create slippage, impact, and execution delay. The apparent alpha may not scale.

Shorting and Hedging Costs

A long–short strategy may show strong spread returns in gross terms, but once we include borrowing fees, margin requirements, and hedge-related costs, the realized return to deployable capital can be much lower.

- **Borrowing cost:** A-shares are often around 8%–11%, Hong Kong stocks around 3%–8% for more liquid names but can exceed 10%, and U.S. easy-to-borrow names can be near 0%–1% while hard-to-borrow names can easily reach double digits.
- **Margin requirements:** For A-share index futures, the CFFEX minimum trading margin for major equity index futures such as the CSI 500 contract is 8% of contract value, while actual broker or futures-company requirements can be higher.
- **Hedging cost:** index futures basis, rollover cost, and margin usage affect the return of hedged portfolios. For example, CSI 500 index futures have recently traded at a meaningful annualized discount; around April 24, 2026, reported annualized discounts were roughly 6%–8% for several IC contracts.
- **Availability constraints:** some stocks may be difficult or impossible to borrow at the required size.

Weak Economic Robustness

Some backtests look convincing statistically, but remain unconvincing economically.

- the signal may have no clear risk-based, behavioral, or institutional explanation;
- the factor may fit historical data but fail to reflect a stable economic mechanism;
- without an economic foundation, it is hard to argue that the relation should persist in the future.

Definition

Weak economic robustness means that the model may be statistically significant in sample, but lacks a credible economic rationale for why the effect should exist and survive out of sample.

Why Lack of Economic Foundation Is Dangerous

When a factor has no convincing economic logic, its performance can disappear as soon as the market environment changes.

- macroeconomic conditions can change the pricing of risk;
- regulation or policy can change incentives and trading behavior;
- market participants may arbitrage away a pattern that has no structural foundation;
- without an economic explanation, it is difficult to know when the strategy should work and when it should fail.

Main idea

Economic intuition does not guarantee success, but the absence of economic intuition makes it much harder to believe that a backtest reflects a durable source of return.

Example 1: The Nifty Fifty

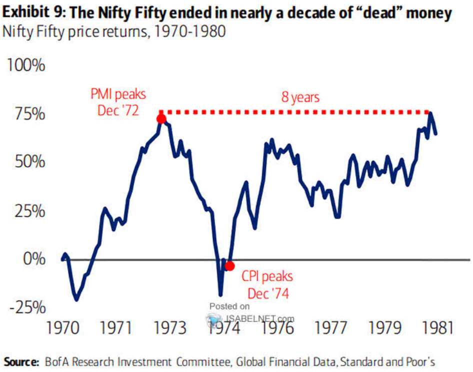
The Nifty Fifty was a group of large U.S. growth stocks that became extremely popular in the late 1960s and early 1970s.

- many investors believed these firms were so strong that valuation no longer mattered;
- the strategy effectively became "buy great companies at any price";
- this looked persuasive for a period, but it lacked a robust economic foundation because it ignored the role of valuation in expected return.

Lesson

When inflation, interest rates, and macro conditions changed in the 1970s, the apparent superiority of the Nifty Fifty collapsed quickly. A strategy can look compelling in one regime, but fail when its economic logic is weak.

The Nifty Fifty: Illustration



The Nifty Fifty episode is often remembered as a period when investors treated a group of strong companies as if valuation no longer mattered. Once the macro regime changed, that belief proved fragile.

Example 2: LTCM

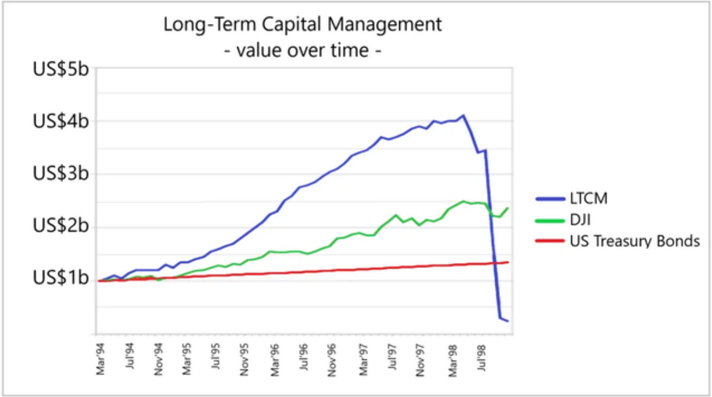
Long-Term Capital Management built highly leveraged convergence trades based on the idea that relative mispricings would revert.

- historically, many spreads had indeed appeared stable and mean-reverting;
- the backtests and statistical models looked strong under normal market conditions;
- however, the strategy relied on the assumption that funding liquidity, market liquidity, and correlation structure would remain sufficiently stable.

Lesson

When market stress and policy conditions changed in 1998, those assumptions broke down. LTCM shows that a statistically impressive model can fail catastrophically if its economic foundation is incomplete and its regime dependence is underestimated.

LTCM: Illustration



LTCM is a classic reminder that a model can look rigorous and highly profitable in normal periods, yet fail abruptly when liquidity, leverage, and market correlations behave differently from the historical sample.

Robustness Checks: A Practical Workflow

After the backtest is implemented correctly, robustness checks ask a narrower question: *how should we test whether the result still survives reasonable changes in specification and environment?*

- **Step 1:** vary key model parameters;
- **Step 2:** re-evaluate the strategy across subperiods and market regimes;
- **Step 3:** stress-test implementation assumptions such as costs, liquidity, and shorting conditions;
- **Step 4:** check whether the result survives alternative universe and sample definitions.

Working principle

Each check should change one dimension at a time, so that we can see clearly which assumption the strategy depends on most heavily.

Robustness Check 1: Parameter Sensitivity

A first test is to vary important parameters and see whether the main conclusion remains similar.

Typical choices to perturb include:

- number of selected assets N;
- rebalance frequency;
- factor lookback window;
- weighting scheme.

How to do it

Run the strategy over a reasonable grid of nearby parameter values and compare whether rank ordering, alpha sign, Sharpe ratio, and drawdown behavior remain broadly consistent.

Robustness Check 2: Subperiod Analysis

A strategy should then be examined across different market environments.

- bull markets;
- bear markets;
- high-volatility and low-volatility periods;
- earlier and later parts of the sample.

How to do it

Split the sample by time or regime, report the same performance metrics in each block, and check whether the strategy remains directionally consistent rather than being driven by one exceptional episode.

Robustness Check 3: Implementation Stress Tests

Implementation assumptions are never known with precision, so they should be stressed explicitly.

Useful stress tests include:

- transaction cost scenarios;
- stricter liquidity or participation caps;
- more conservative shorting or hedging assumptions;
- blocked trades under price limits, suspension, or circuit-breaker conditions.

For example, the strategy can be evaluated under several cost assumptions:

$$c \in \{0, c_1, c_2, c_3\}.$$

How to do it

Re-run the same backtest under progressively more conservative implementation assumptions and observe how quickly net performance deteriorates.

Robustness Check 4: Universe and Sample Alternatives

Finally, the strategy should be tested under reasonable alternative sample definitions.

- use a broader or narrower investable universe;
- exclude very small or illiquid stocks;
- compare index-constituent universes with all-stock universes;
- shift sample start and end dates moderately.

How to do it

The goal is not to search for the best-looking universe, but to verify that the strategy does not depend entirely on one especially favorable sample definition.

Backtest Evaluation Checklist

By the final stage, the object of evaluation is not only a factor but a complete strategy.

- **Return and risk:** annualized return, excess return, Sharpe ratio, and maximum drawdown;
- **Implementation realism:** turnover, transaction costs, liquidity constraints, and shorting or hedging assumptions;
- **Data integrity:** timing alignment, information leakage, and sample construction;
- **Robustness:** parameter sensitivity, subperiod analysis, implementation stress tests, and alternative universes.

Key message

A credible backtest should be judged not only by performance, but also by whether the research process is realistic, disciplined, and robust.

What a Credible Backtest Report Must Contain

A credible report should contain enough detail for others to understand how the result was produced and how fragile it may be.

- a clear strategy definition;
- data source, universe, and sample period;
- timing conventions and implementation assumptions;
- robustness evidence under alternative settings;
- limitations, risks, and likely failure scenarios.

Important

Transparent reporting increases the value of the research even when the final result is modest, because it makes the evidence interpretable rather than merely attractive.

How to Interpret a Positive Backtest

A positive backtest is encouraging, but it should be interpreted carefully.

- it is evidence, not proof;
- its credibility depends on timing discipline, implementation realism, and robustness;
- it may fail when market structure, macro conditions, or policy environments change.

Final message

Good quantitative research is not only about finding profitable historical patterns. It is about testing whether those patterns remain credible after disciplined implementation, robustness checks, and honest interpretation.

Thank you!